

Category	Discipline Specific Core -I			
Course Title	Machine Learning			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	3	0	4	5
Pre-requisites (if any)	Basic Knowledge of Python, Linear Algebra Fundamentals, Basics of Statistics and Probability, Data Structures and DBMS Fundamentals			

Course Objectives:

1	Understand the fundamental concepts, learning paradigms, applications, advantages, limitations, and tools of Machine Learning.
2	Develop knowledge of data preprocessing and feature engineering techniques for improving data quality and model performance..
3	Familiarize students with model training, model evaluation methods, probabilistic foundations, and ensemble learning approaches.
4	Understand and apply supervised learning algorithms for classification and regression problems.
5	Analyze unsupervised learning techniques including clustering methods and association rule learning algorithms for pattern discovery.

1. Course Contents:

Unit	Contents	Weightage
I	Foundations of Machine Learning: Introduction, Human Learning Vs Machine Learning, Types of Machine Learning (Supervised, Unsupervised, Semi Supervised and Reinforcement Learning), Comparison of all learning methods, Problems Not to be Solved using Machine Learning, Applications of Machine Learning, Tools in Machine Learning, Issues and Advantages in Machine Learning	20%
II	Data Preprocessing and Feature Engineering Overview of Data Preprocessing in Machine Learning, Machine Learning Activities, Basic Types of Data in Machine Learning, Data Quality Issues, Data Remediation Techniques (Handling Missing Values, Handling Outliers).	20%

	Introduction to Feature Engineering, Feature Transformation, Encoding Techniques, Feature Subset Selection, Standardization, Dimensionality Reduction, Feature Scaling	
III	Model Training, Evaluation, Probabilistic Foundations and Ensemble Learning Model Introduction, Selecting a Model, Training a Model, Model Representation and Interpretability, Evaluating Performance of a Model, Improving Performance of a Model. Introduction to Probability, Bayes' Theorem Bagging, Boosting, Random Forest	20%
IV	Supervised Learning: Introduction, Example of Supervised Learning, Advantages, Disadvantages and Applications of Supervised Learning Algorithms, Classification Model, Classification Learning Steps, Common Classification Algorithms (K-Nearest Neighbour (KNN), Decision Tree, Support Vector Machines (SVM), Naïve Bayes Classifier, Logistic Regression) Example of Regression, Common Regression Algorithms (Simple Linear Regression, Multiple Linear Regression)	20%
V	Unsupervised Learning: Introduction, Unsupervised Learning versus Supervised Learning, Advantages, Disadvantages and Applications of Unsupervised Learning Algorithms, Clustering and its Types (K-means, K-medoids, Hierarchical, DBSCAN), Finding Pattern using Association Rule, Apriori algorithm for Association Rule Learning.	20%

Unit wise and Topic wise Suggested Free Ai Tool/Platforms

Unit-I

Topic	Suggested Free AI Tool / Platform
Introduction to Machine Learning	ChatGPT
Human Learning Vs Machine Learning	Google Gemini
Types of Machine Learning	ChatGPT
Supervised Learning	Google Teachable Machine
Unsupervised Learning	Orange Data Mining Cloud Demos
Semi-Supervised Learning	Google Colab AI Assistant
Reinforcement Learning	OpenAI Playground
Comparison of Learning Methods	Claude AI

Topic	Suggested Free AI Tool / Platform
Problems Not to be Solved using ML	Perplexity AI
Applications of Machine Learning	Microsoft Copilot
Tools in Machine Learning	Kaggle Learn
Issues in Machine Learning	Perplexity AI
Advantages of Machine Learning	ChatGPT

Unit-II

Topic	Suggested Free AI Tool / Platform
Overview of Data Preprocessing in Machine Learning	Google Colab AI
Machine Learning Activities	ChatGPT Free
Basic Types of Data in Machine Learning	Kaggle Learn
Data Quality Issues	Dataiku Free Edition
Handling Missing Values	KNIME Analytics Platform
Handling Outliers	Orange Data Mining
Introduction to Feature Engineering	RapidMiner Free Studio
Feature Transformation	Google Colab AI
Feature Subset Selection	Weka
Dimensionality Reduction	Teachable Machine
Feature Scaling	Google Colab, Scikit-learn, Pandas, NumPy, Kaggle Notebooks, Orange Data Mining, Weka, Jupyter Notebook, RapidMiner
Standardization	Google Colab, Scikit-learn, Pandas, NumPy, Kaggle Notebooks, Orange Data Mining, Weka, Jupyter Notebook, RapidMiner
Encoding Techniques	Google Colab, Scikit-learn, Pandas, Category Encoders Library, NumPy, Kaggle Notebooks, Orange Data Mining, Weka, Jupyter Notebook

Unit-III

Topic	Suggested Free AI Tool / Platform
Introduction to Machine Learning Models	ChatGPT

Topic	Suggested Free AI Tool / Platform
Selecting a Model	Google Gemini
Training a Model	Google Teachable Machine
Model Representation and Interpretability	ChatGPT
Evaluating Performance of a Model	Microsoft Copilot
Improving Performance of a Model	Claude AI
Introduction to Probability	Wolfram Alpha
Bayes' Theorem	Wolfram Alpha
Bagging	Google Colab, Scikit-learn, Orange Data Mining, Kaggle Notebooks, Weka
Boosting	Google Colab, Scikit-learn, Kaggle Notebooks, XGBoost, LightGBM, CatBoost, Orange Data Mining, Weka, KNIME Analytics Platform
Random Forest	Orange Data Mining, RapidMiner AI Hub

Unit-IV:

Topic	Suggested Free AI Tools / Platforms
Introduction to Supervised Learning	ChatGPT, Google Gemini, Claude AI
Examples of Supervised Learning	ChatGPT, Machine Learning for Kids
Advantages, Disadvantages and Applications	Perplexity AI, Microsoft Copilot
Classification Model	Teachable Machine, Orange Data Mining
Classification Learning Steps	ChatGPT, Google Gemini
K-Nearest Neighbour (KNN)	Orange Data Mining, Weka
Decision Tree	Teachable Machine, Weka
Support Vector Machine (SVM)	Weka, Orange Data Mining
Naïve Bayes Classifier	ChatGPT, Weka
Logistic Regression	Google AI Studio, Orange Data Mining
Regression Examples	ChatGPT, Google Gemini
Simple Linear Regression	Teachable Machine, Orange Data Mining
Multiple Linear Regression	RapidMiner AI Hub, Weka

Unit-V:

Topic	Suggested Free AI Tool / Platform
Introduction to Unsupervised Learning	ChatGPT
Unsupervised Learning versus Supervised Learning	Google Gemini
Advantages and Disadvantages of Unsupervised Learning	Claude AI
Applications of Unsupervised Learning Algorithms	Microsoft Copilot
Clustering and its Types	Perplexity AI
K-Means Clustering	ChatGPT
K-Medoids Clustering	Claude AI
Hierarchical Clustering	Google Gemini
DBSCAN Clustering	Perplexity AI
Finding Patterns using Association Rule	Microsoft Copilot
Apriori Algorithm for Association Rule Learning	ChatGPT

Course Outcomes (COs)

After successful completion of this course, students will be able to:

CO Code	Course Outcome
CO1	Understand fundamental concepts, learning paradigms, applications, advantages, limitations, and issues of Machine Learning.
CO2	Apply data preprocessing and feature engineering techniques including handling missing values, outliers, encoding, scaling, and dimensionality reduction.
CO3	Analyze model training, evaluation, probabilistic concepts, Bayes' theorem, and ensemble learning techniques such as Bagging, Boosting, and Random Forest.
CO4	Implement and compare supervised learning algorithms for classification and regression problems.
CO5	Analyze unsupervised learning techniques including clustering methods and association rule learning algorithms for pattern discovery.

Program Outcomes (POs)

PO Code	Program Outcome
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PO1	Computational Knowledge
PO2	Problem Analysis
PO3	Design / Development of Solutions
PO4	Modern Tool Usage
PO5	Individual and Team Work
PO6	Professional Ethics
PO7	Communication Skills
PO8	Project Management
PO9	Life-long Learning

CO-PO Mapping Matrix

Mapping Scale:

- 3 = High
- 2 = Medium
- 1 = Low

COs \ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9
CO1	3	2	1	1	-	-	1	-	2
CO2	3	3	2	3	1	-	1	-	2
CO3	3	3	2	2	1	-	1	1	2
CO4	3	3	3	3	1	-	1	1	2
CO5	3	3	2	2	1	-	1	-	2

Reference Books:

- 1) Machine Learning By Saikat Dutt, Subramanian Chandramouli, Amit Kumar Das, Pearson Publication
- 2) Machine Learning with Python for Everyone, Mark Fenner, Pearson
- 3) Machine Learning, Anuradha Srinivasaraghavan, Vincy Joseph, Wiley
- 4) Peter Harrington, “Machine Learning in Action”, DreamTech
- 5) Henrik Brink, Joseph Richards, Mark Fetherolf, “Real-World Machine Learning”, DreamTech



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Semester: III Effective From: June 2026 – Batch 2025

- 6) Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, 3rd Edition, 2022.

Webliography:

<https://www.geeksforgeeks.org/machine-learning/>

https://www.tutorialspoint.com/machine_learning_with_python/index.htm



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Tools: Python / Google Co-lab.

Accomplishment of the student after completing the course:

After completion of the course, students will be able to understand the fundamental concepts and applications of Machine Learning, perform data preprocessing and feature engineering techniques, and develop machine learning models using supervised and unsupervised learning algorithms. Students will be capable of applying classification, regression, clustering, and association rule learning techniques to solve real-world problems. They will also be able to evaluate and improve model performance using appropriate evaluation methods and ensemble learning approaches such as bagging, boosting, and Random forest. Furthermore, students will be able to analyze patterns in datasets and interpret the outcomes of machine learning models for decision-making and business applications.